Absolutely. This is where we move from **“what Mawiz Academy should be”** to **“what we actually need to build.”**

MAWIZ ACADEMY

MVP Functional Specification

Product: Mawiz Academy
Company: Mawiz Technologies Ltd
Document: MVP Functional Specification
Version: 1.0
Status: Development Blueprint

---

1. PRODUCT OBJECTIVE

Mawiz Academy's MVP will validate one central hypothesis:

> **Can we help a person move from having a skill to demonstrating practical and workplace readiness?**

The MVP will therefore combine four core systems:

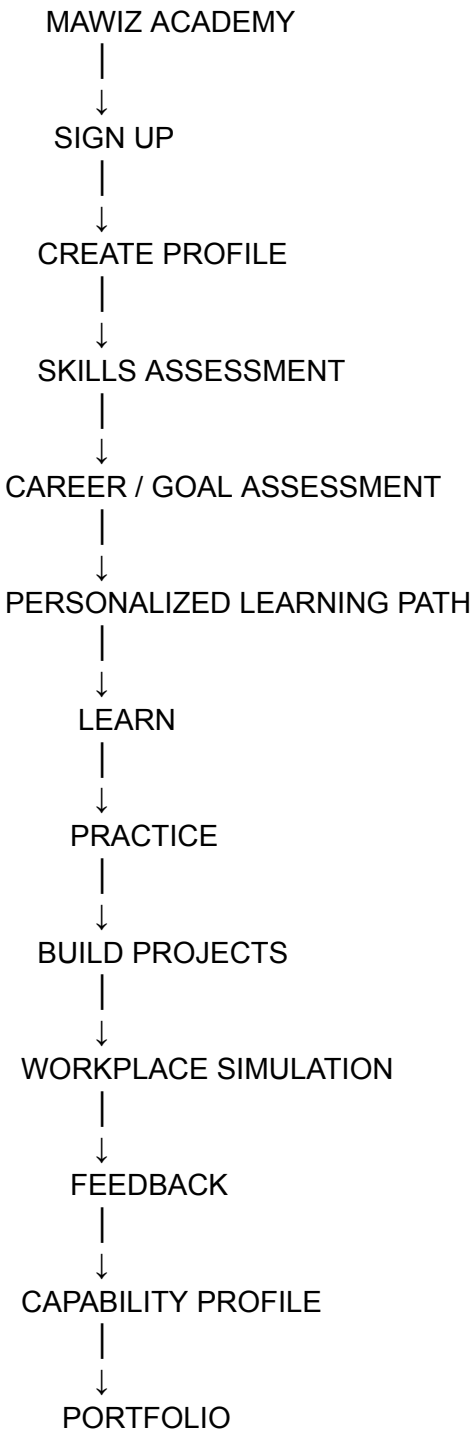
LEARNING
+
PRACTICE
+
WORKPLACE SIMULATION
+
CAPABILITY EVIDENCE

The MVP should **not** attempt to become a complete job marketplace, social network, or massive course marketplace.

The first product must prove the core value proposition.

---

2. MVP CORE USER JOURNEY



---

3. MVP USER TYPES

The system will initially support four roles.

3.1 Learner

The main platform user.

Can:

- create an account;
- complete assessments;
- select career goals;
- enroll in pathways;
- consume lessons;
- complete quizzes;
- submit assignments;
- complete projects;
- participate in simulations;
- receive feedback;
- track progress;
- build a portfolio.

3.2 Instructor

Responsible for learning content.

Can:

- create courses;
- create modules;
- create lessons;
- create quizzes;
- create assignments;
- create projects;
- review learner submissions.

3.3 Mentor / Reviewer

Responsible primarily for practical evaluation.

Can:

- review projects;
- evaluate workplace simulations;
- provide feedback;
- assess competencies;
- approve evidence.

3.4 Administrator

Controls the entire system.

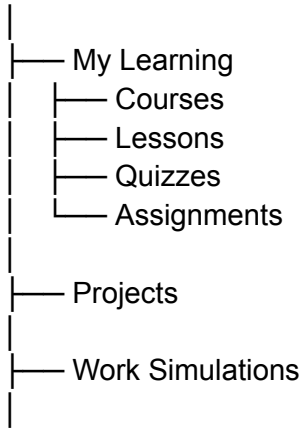
Can:

- manage users;
- manage courses;
- manage assessments;
- manage projects;
- manage simulations;
- manage instructors;
- manage mentors;
- view analytics;
- manage platform settings.

---

4. MVP INFORMATION ARCHITECTURE

Learner Application

Dashboard


├── Skills
│
├── Portfolio
│
├── AI Coach
│
└── Profile

---

5. PUBLIC WEBSITE

The public website should contain:

Home

Core message:

> **Learn the skill. Experience the work. Become work-ready.**

Sections:

- problem;
- solution;
- how Mawiz Academy works;
- learning pathways;
- workplace simulations;
- learner outcomes;
- testimonials;
- pricing;
- CTA.

About

Explain:

- Mawiz Academy;
- Mawiz Technologies;
- mission;
- vision.

Programs

Display available learning pathways.

How It Works

Assess
↓
Learn
↓
Practice
↓
Build
↓
Simulate
↓
Prove

Pricing

Future subscription/cohort pricing.

Login

Register

---

6. AUTHENTICATION SYSTEM

Registration

Required:

- first name;
- last name;
- email;
- password.

Optional:

- phone;
- profile photo.

Future:

- Google authentication;
- LinkedIn authentication.

Login

Fields:

- email;
- password.

Functions:

- login;
- remember session;
- forgot password.

---

7. ONBOARDING

After registration, the learner enters onboarding.

Step 1 — Personal Information

Collect:

- name;
- location;
- education level;
- employment status.

Step 2 — Career Goal

Ask:

> What do you want to become?

Examples:

- Backend Developer
- Frontend Developer

- Data Analyst
- Cybersecurity Analyst
- Digital Marketer
- AI Content Creator
- Automation Specialist

Step 3 — Experience

Ask:

> What is your current experience level?

Options:

- Beginner
- Some knowledge
- Intermediate
- Advanced
- Professional

Step 4 — Skills

Learner selects existing skills.

Step 5 — Assessment

Learner completes initial assessment.

Step 6 — Recommended Path

System generates:

YOUR RECOMMENDED PATH

Career Goal:

Backend Developer

Current Level:
Beginner / Intermediate

Priority Skills:
Node.js
REST APIs
MongoDB
Git

Workplace Skills:
Communication
Documentation
Team Collaboration

Recommended Program:
Backend Developer Work-Ready Path

---

8. LEARNER DASHBOARD

The dashboard is the learner's command center.

Dashboard Components

Welcome

> Good morning, David.

Current Path

Backend Development

Progress

Overall Progress
 78%

Current Task

> Complete REST API Authentication Project

Workplace Challenge

Respond to a client change request.

Skills

Node.js 
APIs 
Git 
Communication 
Documentation 

Recent Achievements

- API Project completed
- Teamwork simulation completed
- Git assessment passed

---

9. COURSE SYSTEM

Courses consist of:

Course
↓
Modules
↓
Lessons
↓
Activities
↓
Assessment

Example:

Backend Development
│
├── Module 1: JavaScript Foundations
├── Module 2: Node.js
├── Module 3: Express
├── Module 4: Databases
├── Module 5: REST APIs
├── Module 6: Authentication

├── Module 7: Git & Collaboration
└── Module 8: Deployment

---

10. LESSON TYPES

The system should support:

Video

Embedded or hosted video.

Text

Rich-text lessons.

Documents

PDF/resources.

Quiz

Multiple-choice and other structured questions.

Interactive Exercise

Future capability.

Assignment

Learner submits an artifact.

---

11. QUIZ ENGINE

Quiz types for MVP:

- multiple choice;
- true/false;
- multiple answer.

Each question can have:

- question;
- options;
- correct answer;
- explanation;
- points.

System automatically calculates scores.

---

12. ASSIGNMENT SYSTEM

An assignment contains:

- title;
- description;
- instructions;
- expected deliverable;
- deadline;
- evaluation criteria.

Example:

> Create a REST API with authentication and document the endpoints.

Submission types:

- text;
- URL;
- file;
- GitHub repository.

---

13. PROJECT SYSTEM

Projects are more important than ordinary assignments.

A project contains:

Project

│
├── Business Context
├── Problem
├── Requirements
├── Deliverables
├── Constraints
├── Deadline
├── Evaluation Criteria
└── Submission

Example:

Project

> Build a customer management API for a fictional company.

Deliverables

- source code;
- API documentation;
- database;
- deployment;
- project explanation.

---

14. PROJECT EVALUATION

Projects should be evaluated against a rubric.

Example:

Competency	Weight
Technical implementation	30%
Problem solving	20%
Code quality	15%
Documentation	10%
Communication	10%

Deadline management	10%
Presentation	5%

The exact weighting should vary by project.

---

15. WORKPLACE READINESS SYSTEM

This is a core differentiator.

The system should contain a dedicated **Workplace Readiness** area.

Initial competencies:

1. Communication
2. Professional Writing
3. Teamwork
4. Time Management
5. Accountability
6. Problem Solving
7. Receiving Feedback
8. Giving Updates
9. Client Communication
10. Workplace Etiquette

---

16. WORKPLACE SIMULATION ENGINE

The simulation engine presents realistic professional scenarios.

Example:

Scenario

> You are a junior backend developer. Your team lead assigns you a task but the requirements are unclear.

Learner must respond.

Possible response:

> "I understand the task. Before I begin, could you clarify whether the API should support authentication and what response format the frontend expects?"

The system evaluates the response against predefined competencies.

---

17. SIMULATION TYPES

Communication Simulation

Learner responds to:

- manager;
- colleague;
- client.

Deadline Simulation

Learner must handle:

- competing tasks;
- deadline pressure;
- delayed work.

Team Simulation

Learner works through:

- disagreement;
- collaboration;
- delegation.

Client Simulation

Learner handles:

- unclear requirements;
- revisions;

- complaints;
- scope changes.

Feedback Simulation

Learner receives criticism and must respond appropriately.

---

18. SIMULATION SCORING

Instead of simply saying:

> 80%

the platform should eventually explain performance.

Example:

Communication Strong
Professionalism Strong
Problem Solving Moderate
Clarification Strong
Escalation Needs Improvement

This creates useful developmental feedback.

---

19. AI ASSISTANT

The AI Assistant should have several modes.

Mode 1 — Learning Tutor

Learner asks:

> Explain REST APIs to me.

AI responds at the learner's level.

Mode 2 — Practice Coach

Learner requests:

> Give me a backend development challenge.

AI generates an appropriate exercise.

Mode 3 — Workplace Coach

Learner asks:

> How should I tell my manager I won't finish the task today?

AI teaches professional communication.

Mode 4 — Interview Coach

AI conducts simulated interviews.

Mode 5 — Career Coach

AI helps learners understand:

- career paths;
- skills required;
- portfolio development;
- professional presentation.

---

20. IMPORTANT AI RULE

The AI should **not simply give learners answers**.

Where appropriate, it should guide the learner through the problem.

For example:

Instead of:

> "Here is the solution."

It can say:

> "Before I give you the solution, let's identify what the error is telling us."

This encourages learning rather than AI dependency.

---

21. AI PERSONALIZATION

The AI should eventually use learner information such as:

Career Goal
Current Skill Level
Completed Lessons
Assessment Results
Project Performance
Simulation Results
Weak Areas
Learning History

to provide recommendations.

Example:

> "You've completed the Node.js module, but your last project showed difficulty with error handling. We recommend completing the error-handling practice before moving to the next project."

---

22. SKILL ENGINE

Every learner should have a structured skill profile.

Example:

Technical Skills
│
├── JavaScript
├── Node.js
├── Express
├── MongoDB
├── Git
└── REST APIs

Professional Skills
│
├── Communication
├── Teamwork
├── Documentation
├── Time Management
└── Problem Solving

Each skill can have a development state:

Not Assessed
↓
Beginner
↓
Developing
↓
Competent
↓
Demonstrated

---

23. CAPABILITY PROFILE

This becomes one of the most important product outputs.

A learner's profile should show:

Identity

- Name
- Professional headline
- Career goal

Skills

Technical + professional.

Evidence

- projects;
- simulations;
- assignments;
- assessments.

Experience

- practical projects;
- team projects;
- client projects.

Feedback

- mentor;
- instructor;
- project reviews.

Portfolio

Links and work samples.

---

24. PORTFOLIO SYSTEM

Learners can add:

- GitHub;
- websites;
- designs;
- videos;
- reports;
- projects;
- certificates.

Each project should include:

Project Name
Problem

Solution
Technology Used
Learner's Role
Outcome
Evidence

---

25. BADGES

The MVP may include simple achievement badges.

Examples:

- First Project
- API Builder
- Team Player
- Communication Pro
- Problem Solver
- Workplace Ready

Badges should represent actual activities rather than merely gamification.

---

26. ADMIN DASHBOARD

Admin homepage:

Total Learners
Active Learners
Courses
Projects
Simulations
Submissions
Completion Rate

---

27. ADMIN COURSE MANAGEMENT

Admin can:

- create;
- edit;
- publish;
- unpublish;
- delete courses.

Course structure:

```
Course
├── Module
│   ├── Lesson
│   ├── Quiz
│   └── Assignment
└── Project
```

---

28. ADMIN SIMULATION MANAGEMENT

Admin can create:

- scenario;
- context;
- learner role;
- task;
- expected competencies;
- evaluation criteria;
- model answers/examples;
- feedback.

---

29. ADMIN ASSESSMENT MANAGEMENT

Admin can create:

- questions;
- answers;
- points;
- skill category;
- difficulty;
- explanations.

---

30. SUBMISSION REVIEW

Instructor/mentor dashboard:

Pending Reviews
↓
Open Submission
↓
Review Evidence
↓
Score Competencies
↓
Write Feedback
↓
Approve / Request Revision

---

31. NOTIFICATION SYSTEM

MVP notifications:

- assignment due;
- project submitted;
- feedback received;
- new course available;
- assessment result;
- simulation available.

Channels:

- in-app;
- email.

Future:

- WhatsApp.

---

32. DATABASE DESIGN

A simplified relational structure:

users
│
├── profiles
├── roles
├── skills
├── learner_skills
├── courses
│   ├── modules
│   │   ├── lessons
│   │   ├── quizzes
│   │   └── assignments
│
├── enrollments
├── lesson_progress
├── quiz_attempts
├── assignments
├── submissions
├── projects
├── project_submissions
├── simulations
├── simulation_attempts
├── competencies
├── competency_results
├── portfolios
├── achievements
├── notifications
└── ai_conversations

---

33. CORE DATABASE ENTITIES

User

id
first_name
last_name
email
password_hash

role
status
created_at
updated_at

Learner Profile

id
user_id
career_goal
education_level
experience_level
bio
location
profile_image
created_at
updated_at

Skill

id
name
category
description

Learner Skill

id
learner_id
skill_id
level
source
evidence_id
updated_at

Course

id
title
description

category
level
thumbnail
status
created_at
updated_at

Module

id
course_id
title
description
order

Lesson

id
module_id
title
content
type
order

Enrollment

id
learner_id
course_id
status
progress
started_at
completed_at

Project

id
course_id
title
description

requirements
deadline
difficulty

Project Submission

id
project_id
learner_id
submission_url
description
status
score
feedback
submitted_at
reviewed_at

Workplace Simulation

id
title
category
scenario
instructions
difficulty
evaluation_rules

Simulation Attempt

id
simulation_id
learner_id
response
score
feedback
completed_at

Competency

id

name
category
description

Competency Result

id
learner_id
competency_id
level
score
evidence_source
reviewer_id
created_at

---

34. API ARCHITECTURE

A REST API is sufficient for MVP.

Example:

/api/auth
/api/users
/api/profiles
/api/skills
/api/courses
/api/modules
/api/lessons
/api/enrollments
/api/quizzes
/api/assignments
/api/projects
/api/submissions
/api/simulations
/api/competencies
/api/portfolio
/api/achievements
/api/notifications
/api/ai

---

35. AUTHENTICATION API

POST /api/auth/register
POST /api/auth/login
POST /api/auth/logout
POST /api/auth/forgot-password
POST /api/auth/reset-password
GET /api/auth/me

Authentication should use secure token/session management.

---

36. COURSE API

GET   /api/courses
GET   /api/courses/:id
POST   /api/courses
PATCH /api/courses/:id
DELETE /api/courses/:id

---

37. LEARNING API

POST /api/enrollments
GET /api/enrollments
GET /api/enrollments/:id
POST /api/lessons/:id/complete
GET /api/learners/:id/progress

---

38. PROJECT API

GET /api/projects
GET /api/projects/:id
POST /api/projects/:id/submit
GET /api/submissions/:id
POST /api/submissions/:id/review

---

39. SIMULATION API

GET /api/simulations
GET /api/simulations/:id
POST /api/simulations/:id/start
POST /api/simulations/:id/submit
GET /api/simulations/:id/result

---

40. AI API

POST /api/ai/chat
POST /api/ai/tutor
POST /api/ai/practice
POST /api/ai/workplace-coach
POST /api/ai/interview
POST /api/ai/recommend

The backend should control AI access rather than exposing provider API keys to the frontend.

---

41. RECOMMENDED TECH STACK

Given Mawiz Technologies' existing technical direction, the MVP can use:

Frontend

React / Next.js

or a simpler React frontend depending on development resources.

Backend

Node.js + Express.js

This aligns well with your existing backend experience.

Database

PostgreSQL

Recommended for the core Academy because the product contains highly relational data:

- learners;
- courses;
- skills;
- competencies;
- projects;
- assessments;
- submissions.

MongoDB could work, but PostgreSQL gives the MVP a strong relational foundation.

Authentication

JWT/session-based authentication.

Storage

Object storage for:

- profile images;
- certificates;
- submitted files;
- project evidence.

Potential:

- AWS S3.

AI

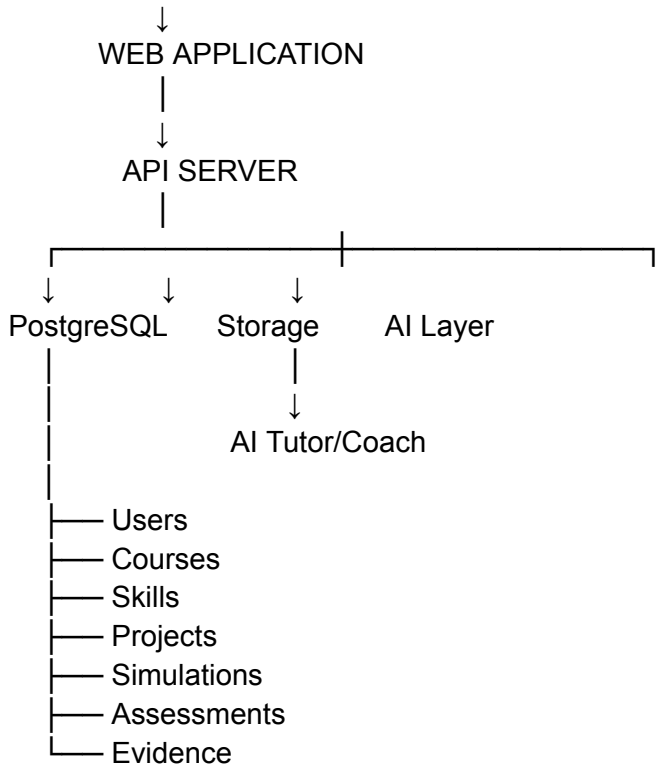
LLM provider through a backend abstraction layer.

The application should avoid tightly coupling itself to a single AI provider.

---

42. HIGH-LEVEL SYSTEM ARCHITECTURE





---

43. AI ARCHITECTURE

Do not make one giant AI prompt responsible for the whole platform.

Instead create specialized AI services.

AI SERVICE
- Tutor
- Practice Generator
- Workplace Coach
- Interview Coach
- Career Coach
- Recommendation Engine
- Assessment Assistant

Each service should have controlled inputs and outputs.

---

44. AI KNOWLEDGE SYSTEM

The AI should have access to approved Mawiz Academy content.

For example:

Course Content
Learning Objectives
Assessment Rubrics
Workplace Framework
Simulation Rules
Career Framework

This reduces hallucination and keeps AI responses aligned with the Academy's curriculum.

A future RAG system can be introduced for this.

---

45. WORKPLACE COMPETENCY ENGINE

This should eventually become a proprietary component of Mawiz Academy.

Example competency:

Professional Communication

Measured through:

- written communication;
- clarity;
- tone;
- structure;
- appropriate escalation;
- response to feedback.

Another:

Accountability

Measured through:

- deadline handling;
- progress updates;
- ownership;
- transparency;
- escalation.

This competency framework could become one of Mawiz Academy's long-term intellectual assets.

---

46. LEARNER EVIDENCE MODEL

Every significant learner action can potentially create evidence.

LESSON
↓
QUIZ
↓
ASSIGNMENT
↓
PROJECT
↓
SIMULATION
↓
FEEDBACK
↓
COMPETENCY EVIDENCE

The platform should preserve this evidence.

This eventually enables the learner's capability profile.

---

47. EXAMPLE CAPABILITY PROFILE

DAVID JOHNSON
Backend Developer

TECHNICAL CAPABILITIES

Node.js

 Demonstrated

REST APIs
 Demonstrated

MongoDB
 Developing

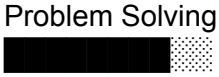
Git
 Demonstrated

WORKPLACE CAPABILITIES

Communication
 Demonstrated

Teamwork
 Developing

Documentation
 Needs Practice

Problem Solving
 Demonstrated

EVIDENCE

Projects: 6
Simulations: 12
Team Projects: 3
Mentor Reviews: 4

The actual UI should avoid implying scientific precision where assessment methods cannot justify it. Evidence and competency levels should be clearly defined.

---

48. MVP SCREEN LIST

Public

1. Home
2. About
3. Programs
4. Program Details
5. Pricing
6. Login
7. Register

Learner

8. Onboarding
9. Assessment
10. Dashboard
11. My Learning
12. Course Details
13. Lesson
14. Quiz
15. Assignment
16. Projects
17. Project Details
18. Submission
19. Workplace Readiness
20. Simulation
21. Simulation Result
22. Skills
23. Portfolio
24. Achievements
25. AI Coach
26. Profile
27. Settings

Admin

28. Admin Dashboard
29. Users
30. Courses
31. Modules
32. Lessons
33. Assessments
34. Projects
35. Simulations
36. Submissions
37. Competencies

38. Analytics
39. Settings

Approximately **39 screens**, although several can initially share layouts/components.

---

49. MVP DEVELOPMENT PRIORITY

P0 — Absolutely Required

- Authentication
- Learner profile
- Course system
- Learning dashboard
- Lessons
- Quizzes
- Assignments
- Projects
- Submission system
- Workplace simulations
- Basic competency tracking
- Admin content management

P1 — Important

- AI tutor
- AI workplace coach
- Portfolio
- Achievements
- Notifications
- Mentor review

P2 — Later

- Employer portal
- Talent search
- Job matching
- Internship matching
- Freelance marketplace
- Social community

- Advanced AI assessment
- Advanced analytics

---

50. MVP ROADMAP

Sprint 1 — Product Foundation

Build:

- project repository;
- database;
- authentication;
- roles;
- basic UI system.

Sprint 2 — Learner System

Build:

- profile;
- onboarding;
- learner dashboard;
- skills.

Sprint 3 — LMS

Build:

- courses;
- modules;
- lessons;
- progress;
- quizzes.

Sprint 4 — Practical Learning

Build:

- assignments;
- projects;
- submissions;
- reviewer feedback.

Sprint 5 — Workplace Readiness

Build:

- competency framework;
- workplace curriculum;
- scenarios;
- simulations;
- results.

Sprint 6 — AI

Build:

- AI tutor;
- AI workplace coach;
- contextual AI chat;
- basic recommendations.

Sprint 7 — Evidence

Build:

- portfolio;
- project evidence;
- competency profile;
- achievements.

Sprint 8 — Testing & Pilot

Recruit an initial group of learners.

Measure:

- completion;
- engagement;
- skill improvement;
- project quality;
- workplace-readiness improvement;
- learner feedback.

---

51. PILOT PROGRAM

Before attempting a large public launch, run a controlled pilot.

Example

25–50 learners

Across perhaps:

- Backend Development
- AI Content Creation
- Data Analysis

Each learner should complete:

Initial Assessment
↓
Learning
↓
Practical Project
↓
Workplace Training
↓
Simulation
↓

Final Assessment

Compare the learner's initial and final capability.

---

52. PILOT SUCCESS CRITERIA

The pilot should answer:

Learning

Did learners actually improve?

Practical

Can learners produce meaningful work?

Workplace

Did their professional competencies improve?

Experience

Did simulations help them understand workplace expectations?

Product

Would learners use the platform again?

Business

Would learners pay for it?

These answers are more important than simply achieving high signup numbers.

---

53. ANALYTICS EVENTS

The platform should track events such as:

user_registered
onboarding_completed
assessment_started
assessment_completed
course_enrolled
lesson_started
lesson_completed
quiz_completed
assignment_submitted
project_started
project_submitted
simulation_started
simulation_completed
feedback_received
portfolio_created
ai_session_started
path_completed

This allows the team to identify where learners drop off.

---

54. SECURITY REQUIREMENTS

The MVP must include:

- password hashing;
- secure authentication;
- authorization;
- role-based access;
- input validation;
- API rate limiting;
- secure file uploads;
- protection against common web vulnerabilities;
- secure environment variables;
- audit logging for sensitive admin actions.

AI API keys must never be exposed to the browser.

---

55. PERFORMANCE REQUIREMENTS

Initial targets:

- dashboard should load quickly on normal mobile internet;
- API responses should generally remain below a few seconds;
- video should use optimized streaming/hosting;
- large project files should use object storage rather than the application server.

Nigeria/mobile-first usage should be considered in the product design.

---

56. MOBILE-FIRST REQUIREMENT

A significant portion of the target market may access the Academy through smartphones.

Therefore:

- responsive web design is mandatory;
- low-bandwidth pages;
- compressed images;
- efficient API responses;
- mobile-friendly forms;
- accessible navigation.

A native mobile app is **not required for MVP**.

---

57. FUTURE EMPLOYER SYSTEM

After validating the learner side, introduce employers.

Employer could:

1. Create account.
2. Define talent requirements.
3. Search candidate profiles.
4. Filter by demonstrated competencies.
5. Review projects.
6. Review evidence.
7. Contact candidates.

Eventually:

Employer Requirement
↓
Competency Matching
↓
Verified Evidence
↓
Candidate Shortlist

---

58. FUTURE REAL-WORLD PROJECT MARKETPLACE

A major future opportunity is allowing businesses to submit real problems.

Example:

A small business needs:

> "A customer-management dashboard."

Mawiz Academy converts the problem into a supervised practical project.

Learners work on it.

The business receives a solution.

Learners receive:

- real experience;
- portfolio evidence;
- feedback;
- potentially paid opportunities.

This creates:

> **Learning → Work → Evidence → Opportunity**

---

59. LONG-TERM PRODUCT LOOP

The mature Mawiz Academy could eventually operate as:

LEARNER
↓
ASSESSMENT
↓
LEARNING
↓
PRACTICE
↓
PROJECT
↓
WORK SIMULATION
↓
FEEDBACK
↓
CAPABILITY
↓
PORTFOLIO
↓
OPPORTUNITY
↓
REAL WORK
↓
EXPERIENCE
↓
MORE PROOF
↓
BETTER OPPORTUNITIES

This creates a powerful feedback loop.

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60. THE MOST IMPORTANT PRODUCT DECISION

Mawiz Academy should not make the mistake of thinking:

> **"We need thousands of courses."**

The goal should instead be:

> **"We need a system that produces demonstrably capable people."**

Ten excellent learning pathways with strong practical and workplace components are more useful for validating the concept than hundreds of shallow courses.

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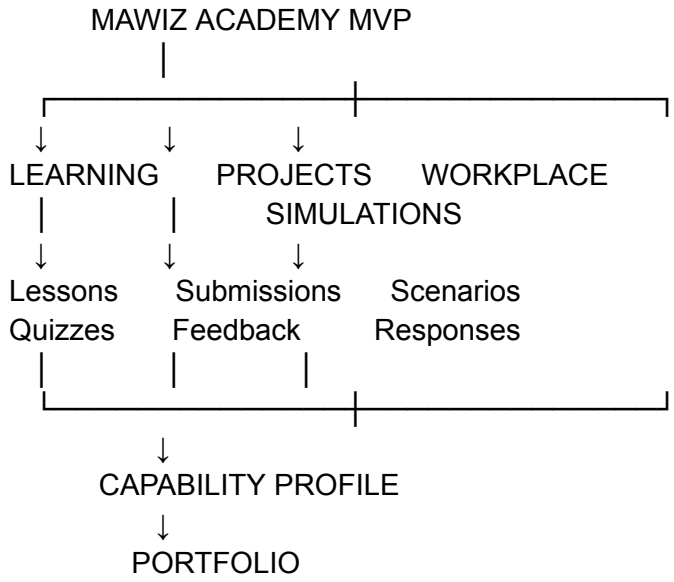
61. MVP DEFINITION IN ONE SENTENCE

> **Mawiz Academy MVP is an AI-assisted learning platform where emerging professionals learn a skill, practice it, complete realistic projects and workplace simulations, receive feedback, and build evidence of their readiness for professional work.**

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62. THE FIRST VERSION WE SHOULD ACTUALLY BUILD

If development resources are limited, I would reduce the first build to this:



AI sits across the system as an **assistant and coach**, rather than being the product itself.

That keeps the MVP focused.

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63. FINAL PRODUCT PRINCIPLE

Every feature should pass this test:

> **Does this move the learner closer to being able to perform meaningful work in the real world?**

If yes:

Build it.

If it is merely a nice-to-have:

Defer it.

Mawiz Academy's competitive opportunity is not simply teaching people more information.

It is helping them cross the difficult bridge between:

> **"I learned it."**

and

> **"I can actually work."**

That bridge should be the heart of the entire product.